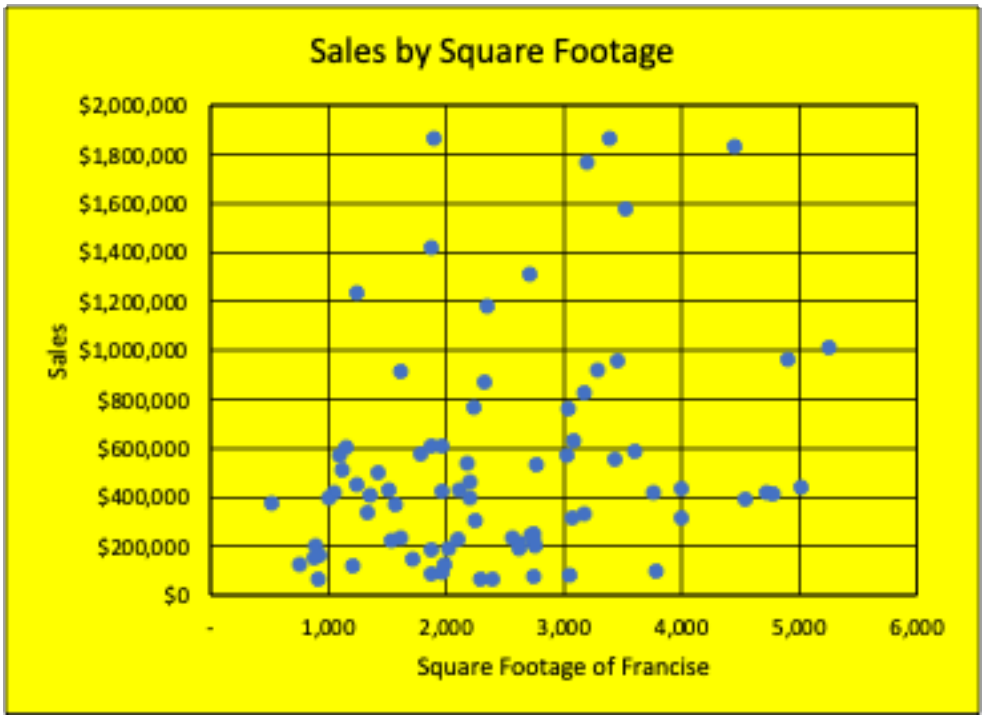
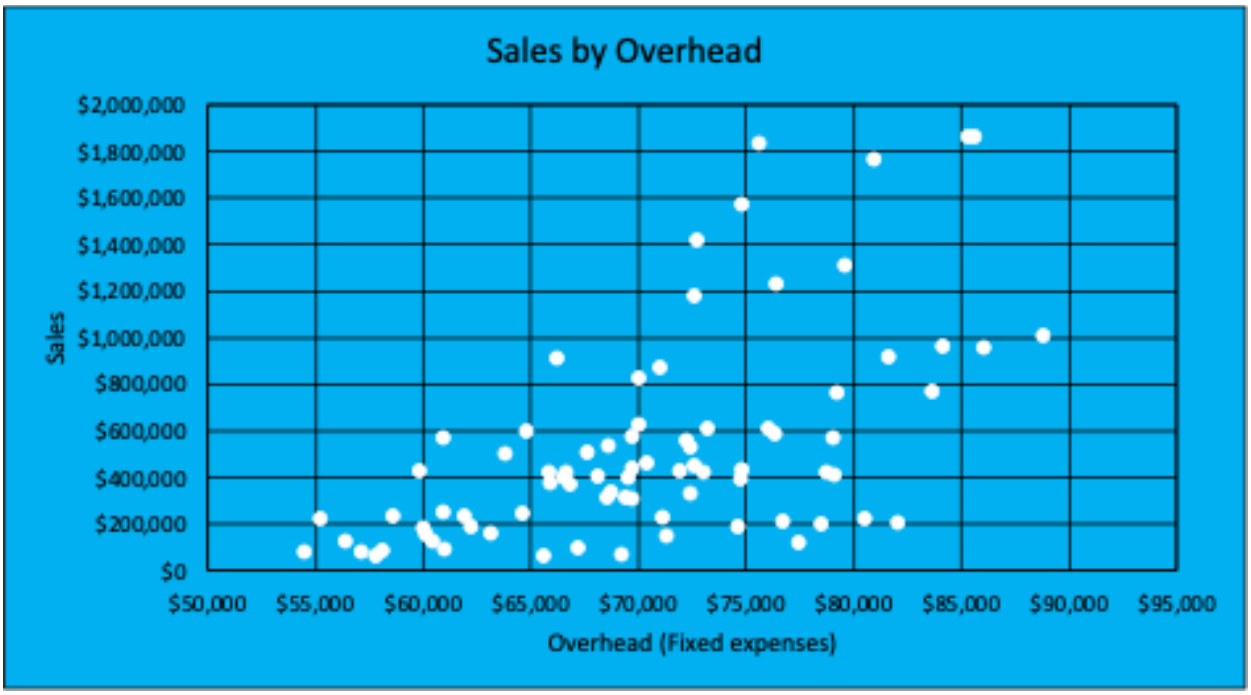

Sales by Square Footage
$2,000,000
$1,800,000
$1,600,000
$1,400,000
$1,200,000
$1,000,000
$800,000
$600,000
$400,000
$200,000
$0
Sales
-
1,000
2,000
3,000
4,000
5,000
6,000
Square Footage of Francise

Sales by Number of Employees
$2,000,000
$1,800,000
$1,600,000
$1,400,000
$1,200,000
$1,000,000
$800,000
$600,000
$400,000
$200,000
$0
Sales
30
40
50
60
70
80
90
Employees



SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.58174642
R Square	0.3384289
Adjusted R S	0.32994722
Standard Err	365.251523
Observations	80

Sales by Overhead
in thousands of dollars

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	5323160.82	5323160.82	39.9011595	1.5213E-08
Residual	78	10405876.7	133408.675		
Total	79	15729037.5			

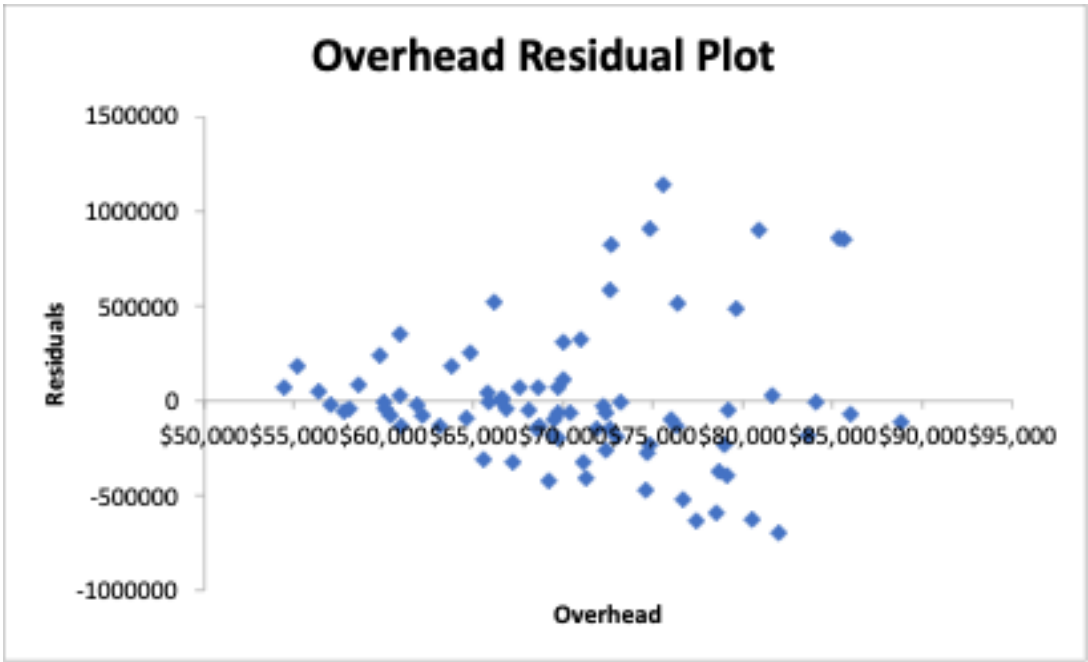
	Coefficients	*Standard Erro*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	-1741.5693	362.058095	-4.8101931	7.2129E-06	-2462.3716	-1020.767
Overhead ($C	32.2496579	5.10543033	6.31673646	1.5213E-08	22.0855269	42.4137889

Regression equation: Sales = -1741.5693 + 32.2496579 * Overhead in thousands of dollars

$$\mathrm{H}_0: \beta = 0$$
$$H_a: \beta \neq 0$$

p-value: 0.000000015213
t statistic: 6.31673646
95% confidence
upper: 42.4137889
lower: 22.0855269
Based on the extremely low p-value, which is approximately 0, we reject the null hypothesis, we have convincing evidence of a useful linear relationship between sales and overhead.

We are 95% confident that the mean change in sales associated with an increase of 1 dollar of overhead is in this interval. (22.0855269,42.4137889)



The points appear to be varying further from the regression line as overhead increases, so it is not appropriate to use the linear regression.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.58954056
R Square	0.34755807
Adjusted R S	0.33061153
Standard Erro	365.070417
Observations	80

linear quadratic regression

ANOVA

	df	SS	MS	F	Significance F
Regression	2	5466753.95	2733376.97	20.5090832	7.2421E-08
Residual	77	10262283.5	133276.41		
Total	79	15729037.5			

	Coefficients	Standard Erro	t Stat	P-value	Lower 95%	Upper 95%
Intercept	1042.42136	2706.41899	0.38516629	0.70117614	-4346.7473	6431.59007
Overhead ($C	-47.397094	76.9017335	-0.6163332	0.5394925	-200.52801	105.733821
overhead qua	0.56236863	0.5417899	1.03798285	0.30252664	-0.5164729	1.64121016

Sales = 1042.42136 + -47.397094 * overhead + 0.56236863 * overhead quadratic

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.57434888
R Square	0.32987664
Adjusted R S	0.32128531
Standard Erro	367.604782
Observations	80

ANOVA

	df	SS	MS	F	Significance F
Regression	1	5188642	5188642	38.3964792	2.5406E-08
Residual	78	10540395.5	135133.275		
Total	79	15729037.5			

	Coefficients	Standard Erro	t Stat	P-value	Lower 95%	Upper 95%
Intercept	-8929.4079	1527.26766	-5.8466555	1.1102E-07	-11969.964	-5888.8515
Overhead ($C	2226.68007	359.345426	6.19648926	2.5406E-08	1511.27829	2942.08186

Sales = -8929.4079 + 2226.68007 * LN(overhead)

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.60123962
R Square	0.36148907
Adjusted R S	0.3362847
Standard Erro	363.520112
Observations	80

ANOVA

	df	SS	MS	F	Significance F
Regression	3	5685875.21	1895291.74	14.3423125	1.7044E-07
Residual	76	10043162.3	132146.872		
Total	79	15729037.5			

	Coefficients	Standard Erro	t Stat	P-value	Lower 95%	Upper 95%	Lower 90.0%	Upper 90.0%
Intercept	-1036.9515	557.469716	-1.8601037	0.06673994	-2147.2486	73.3456911	-1965.2229	-108.68002
Sqr Foot (000	9.49674503	40.2528442	0.2359273	0.81412372	-70.673742	89.6672323	-57.530333	76.523823
Employees	-8.123517	4.98191758	-1.6306004	0.10711233	-18.045866	1.79883181	-16.419164	0.17212982
Overhead ($0	29.1153491	5.80227972	5.01791545	3.3425E-06	17.5591075	40.6715907	19.4536751	38.777023

Sales = intercept + Sqr foot * var + Employees * var + overhead * var

Sales = -1036.9515 + 9.49674503 * sqr foot + -8.123517 * employees + 29.1153491 * Overhead

Sqr foot p value higher than 0.05 so cannot be 95% sure.
Employees p value higher than 0.05 so cannot be 95% sure.
Overhead p value lower than 0.05 which is statistically significant can be 95% confident.

	R squared	coefficents significant
2a	0.3384289	1 out of 1
3a	0.34755807	0 out of 2
3b	0.32987664	1 out of 1
4a	0.36148907	1 out of 3

The best model would be 2a as it has the highest R squared value while having the highest percentage of coefficients significant. 3a is the worst as none of the coefficients are significant even though it has a higher R squared than 2a. They all have relatively low correlation but the best ones have coefficients that have a high p value that explains the relationship.